

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2022-2023

Comprehensive Examination
(EC-3 Makeup)

Course No. : DSECLZG522
Course Title : Big Data Systems
Nature of Exam : Open Book
Weightage : 40%
Duration : 2.5 Hours
Date of Exam : 28-October-2023 (FN)

No. of Pages	= 2
No. of Questions	= 8

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made, if any, should be stated clearly at the beginning of your answer.

Q1. What is the role of the MapReduce programming paradigm in big data analytics, and how does it implement the divide-and-conquer strategy? What are the different types of parallelism used in MapReduce computing paradigm?

(4 Marks)

Q2. You are given access to 10 Terabyte of data with details of daily sales transactions belonging to an e-commerce company. The transactional data files are stored in a folder on HDFS with a structure as shown below:

Transaction No. (Unique)	Invoice	Stock Code	Description	Quantity	Invoice Date	Price(\$)	Customer ID	Country
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It is not required to modify the data. You are not allowed to copy or move the data into any other storage location. Select a suitable analytics tool from the Hadoop ecosystem family to run queries on this data. Give justifications for your selection. Using the tool selected by you, write a query to find the total sales revenue received in \$ from each country.

(6 Marks)

Q3. What is meant by Tunable consistency in NoSQL databases? What is the effect of Write and Read latency on the consistency levels. What consistency levels are recommended by you for (a) Write intensive applications, and for (b) Read intensive applications?

(4 Marks)

Q4. Explain the role of Apache ZooKeeper in the Hadoop ecosystem. How does ZooKeeper ensure coordination and synchronization among distributed components like HBase, HDFS, and YARN? Provide a real-world scenario where ZooKeeper's coordination is critical.

(6 Marks)

Q5. What are the supported storage levels for RDDs? Explain the scenarios in which these storage levels are used? How do you recommend the right storage level for RDDs in your Spark application?

(5 Marks)

Q6. What are the capabilities of Spark SQL ?

The "bank.csv" file has customer data with the following schema:

AcctNo	age	job	Education	balance	loan	name	contact
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Read data from this file into a dataframe and create a table view. Write Spark SQL queries to:

- Find the total number of customers within each job category.
- Find age, education, balance of customers with admin job.

(5 Marks)

Q7. You are working on a streaming data processing application using Spark Streaming. You receive a stream of JSON messages with the following structure: {"user_id": 123, "event_type": "click", "timestamp": "2023-09-01 10:30:45"}. You want to count the number of "click" events in a 5-minute window. Write a Spark Streaming code snippet (in Python or Scala) to accomplish this task.

(5 Marks)

Q8. You are managing a data-intensive project on AWS and need to choose appropriate storage services for different data types. Describe when you would use Amazon EFS (Elastic File System), Amazon EBS (Elastic Block Store), and Amazon S3 (Simple Storage Service) for storing data. Provide examples of data types and use cases for each service.

(5 Marks)
